

Optimal Loan Limit Increases under Profitability–Risk Trade-offs

A Predict-then-Optimize Study on Synthetic East-African Unsecured Lending Data

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Companion code: `loan_limit_optimization.ipynb` · `loan_limit_engine.py` · `ml_behavioral_models.py`

Data note. All borrower-level inputs in this study are synthetic. No proprietary lender portfolios, rates, or performance figures are used or claimed.

Abstract

This report develops a transparent predict-then-optimize framework for retail loan-limit increases. Supervised models estimate PD, offer acceptance, and utilization on an out-of-time-style holdout; a mixed-integer linear program (MILP) allocates discrete increase tiers across risk cohorts to maximize expected profit subject to budget, expected-loss (EL), concentration, and stress-loss proxies. On a synthetic East-African unsecured book (N=4,000), the constrained MILP lifts 12-month expected profit by about 149% versus a no-increase baseline while reducing the exposure-weighted PD proxy (12.9% → 11.0%) and EL/EAD (8.4% → 7.2%). Unconstrained bandit-style heuristics can post higher raw profit but with worse default rates—showing why portfolio constraints belong in the decision layer. Macro stress compresses attainable profit and motivates scenario-conditioned policies. Limitations are discussed explicitly.

1. Introduction

Credit limit increases deepen relationships and interest income but expand EAD and may worsen leverage-driven default risk. In emerging-market digital lending, thin files and macro sensitivity make naive rules fragile. This project answers five assessment questions with an end-to-end reproducible pipeline: stated assumptions, OOT-validated ML predictors, cohort MILP, heuristic benchmarks under macro stress, and operational recommendations.

2. Related context

The problem intersects credit scoring (PD/LGD/EAD), customer response / limit elasticity, constrained portfolio optimization (EL caps, CVaR/stress proxies), and predict-then-optimize designs. Sequential methods (MDP/RL) are discussed as extensions requiring offline evaluation and safety constraints. I privilege interpretability (logistic PD + auditable MILP), which fits regulated lending practice and MSc business-analytics learning outcomes.

3. Data and assumptions

Because live portfolio data are unavailable for this case study, I use a synthetic East-Africa digital unsecured analogue with KES-equivalent units: 4,000 borrowers aggregated into 77 cohorts by PD band × utilization × tenure × product (personal vs salary_advance). Horizon 12 months. Illustrative economics: $r=36\%$ interest, fees=2%, LGD=65%, cost of capital=12%, incremental budget $B=\text{KES } 50,000,000$, max EL/EAD=9%. Tiers $T=\{0\%, 10\%, 25\%, 50\%, 100\%\}$.

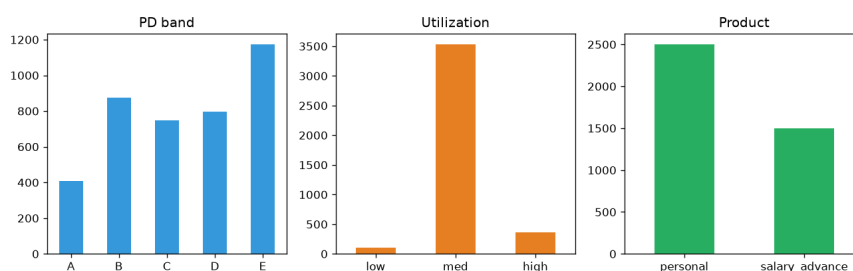


Figure 1. Synthetic portfolio composition.

4. Methodology

4.1 Machine learning (predict)

Simulated historical offers; vintage/tenure-sorted 70/30 out-of-time-style split. Models: logistic regression vs GBM for PD; logistic regression for acceptance; GBM utilization among acceptors. Metrics: AUC, Brier, AP; MAE/R². Interpretability via permutation importance and calibration. Prefer logistic PD when OOT AUC is within ~0.02 of GBM.

Model	Metric	Value
PD logistic	AUC / Brier	0.732 / 0.182
PD GBM	AUC / Brier	0.702 / 0.119

Acceptance logit	AUC	0.762
Utilization GBM	MAE / R ²	0.025 / 0.917

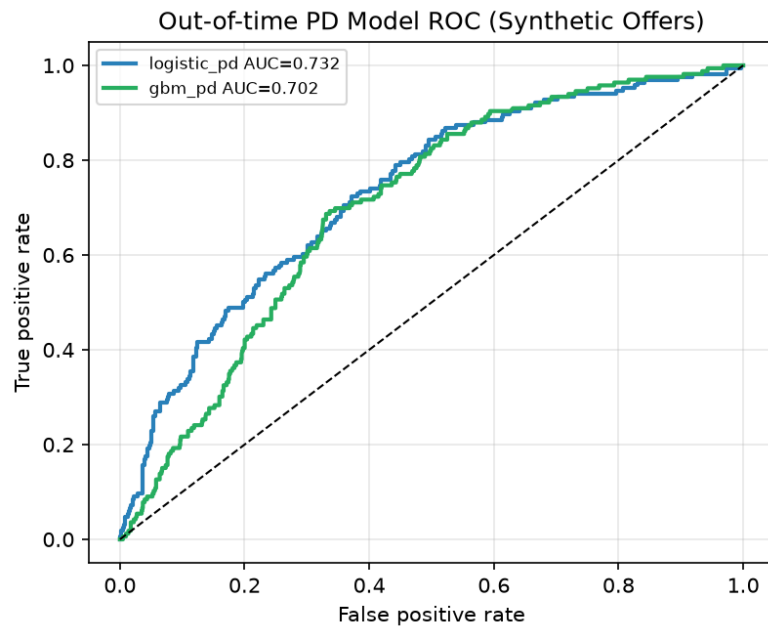


Figure 2. PD ROC (out-of-time).

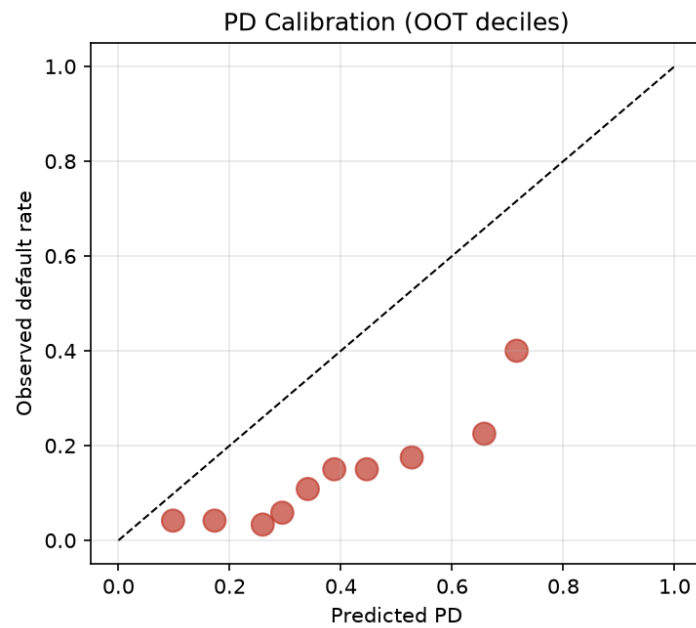


Figure 3. PD calibration.

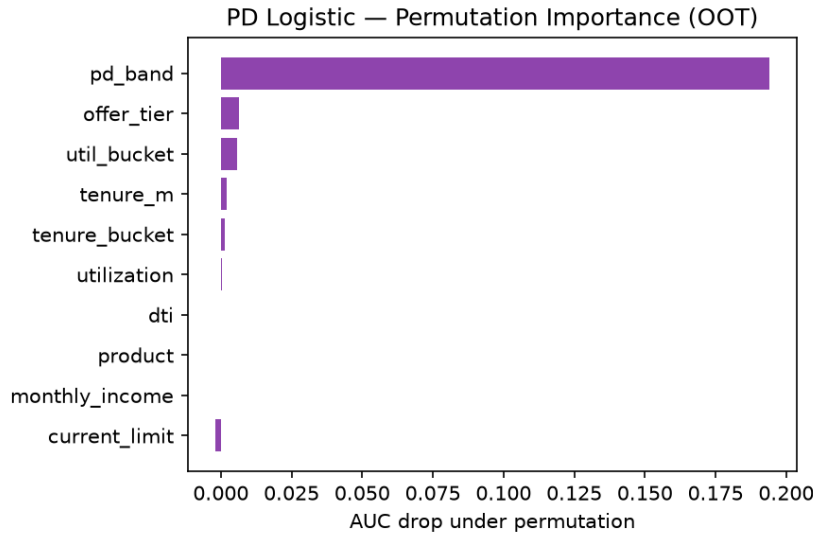


Figure 4. PD permutation importance.

4.2 Behavioral responses & profit

Acceptance rises with tier size and utilization and falls with PD; utilization response has diminishing marginal draw; PD uplifts with leverage when accepted; churn rises when high-util accounts are denied. Account profit (12m) mixes interest, fees, expected loss, capital, opex, and churn cost over accept/decline paths.

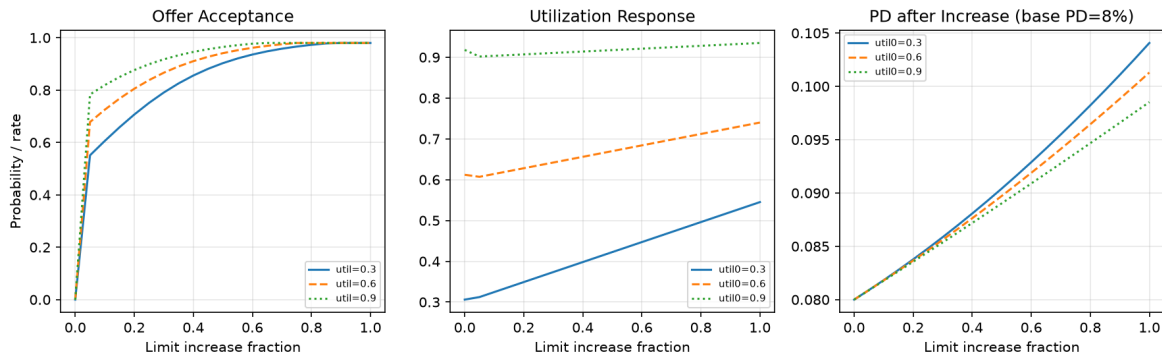


Figure 5. Acceptance, utilization, and PD response curves.

4.3 MILP formulation

Decision $x_{c,t} \in \{0,1\}$: assign cohort c to tier t . Maximize $\sum \Pi_{c,t} x_{c,t}$ subject to one tier per cohort, incremental-limit budget B , $EL \leq \rho \cdot EAD$, high-risk concentration, and a stress-EL proxy. Cohort economics are precomputed (lookup linearization) so the MIP remains tractable and auditable (CBC).

4.4 Benchmarks & macros

Strategies: no_increase, uniform_25, risk_based, greedy_high_util, bandit_heuristic, optimized_milp. Macros: baseline; mild recession (PD $\times$ 1.35, LGD $\times$ 1.10, demand $\times$ 0.92); severe (PD $\times$ 1.90, LGD $\times$ 1.25, demand $\times$ 0.80).

5. Results

5.1 Baseline strategy comparison

Strategy	Profit (KES m)	Lift %	PD proxy %	EL/EAD %	Churn
no_increase	4.11	0.0	12.89	8.38	0.254
uniform_25	5.88	43.1	13.37	8.71	0.107
risk_based	5.73	39.5	11.97	7.78	0.160
greedy_high_util	5.07	23.3	12.26	7.97	0.209
bandit_heuristic	11.82	187.6	16.07	10.47	0.053
optimized_milp	10.25	149.5	11.04	7.19	0.148

Insight. MILP status=Optimal. Profit lift $\approx$ 149.5% with PD proxy 11.04% (vs 12.89% baseline). The bandit posts higher raw profit but worse default metrics—an unconstrained local envelope, not a deployable policy under the stated risk appetite.

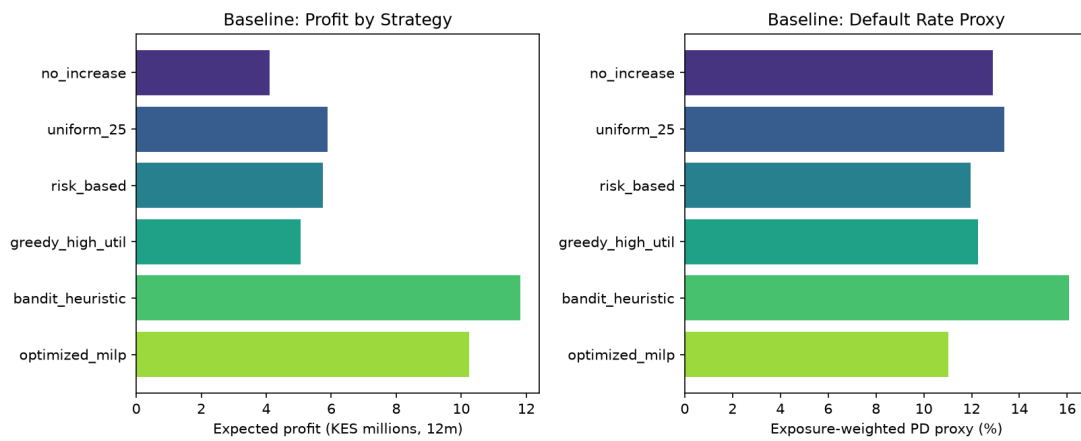


Figure 6. Baseline strategy comparison.

Optimized MILP: Mean Limit Increase Fraction

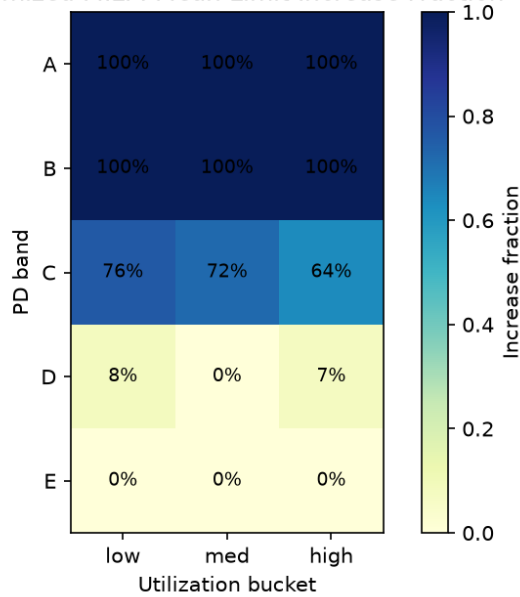


Figure 7. Optimized MILP: mean increase fraction by PD band × utilization.

5.2 Macro stress & frontier

Under mild and severe macros, expected profit falls for all strategies. Stress-aware constraints and scenario playbooks are therefore first-class design requirements for emerging-market books. Varying the EL/EAD ceiling traces a risk–return curve for the MILP.

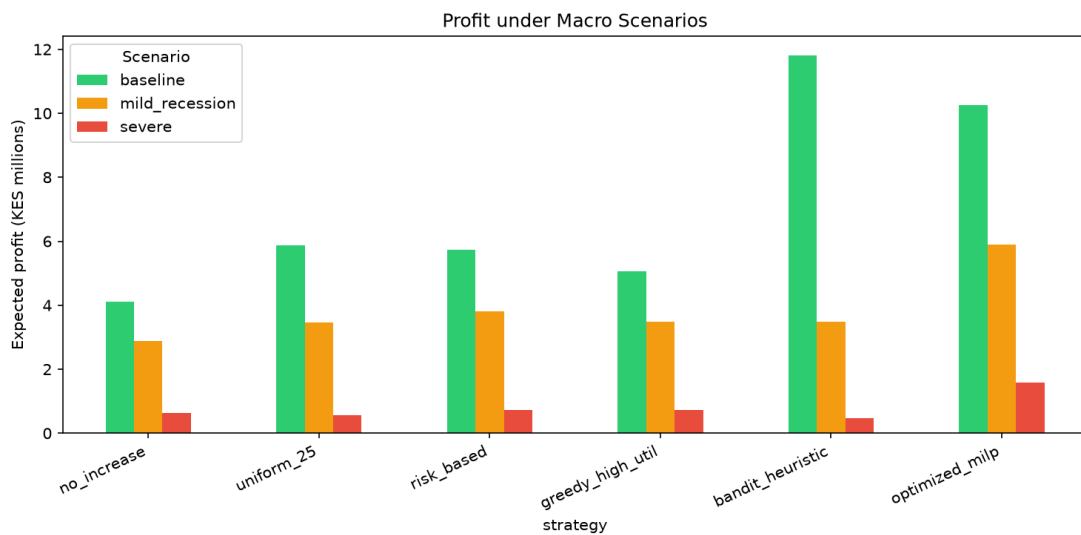


Figure 8. Profit under macro scenarios.

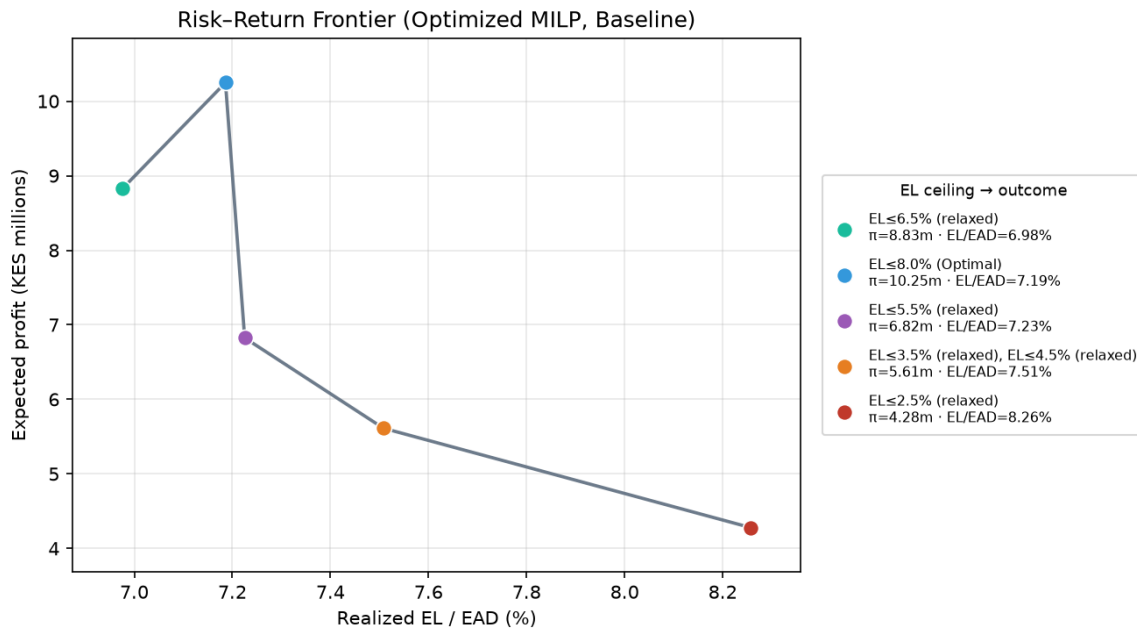


Figure 9. Risk–return frontier (optimized MILP, baseline).

6. Discussion

- Q1 Strategy: Predict-then-optimize with discrete tiers and hard risk/budget constraints; concentrate increases on low-PD × high-util × longer-tenure cohorts.
- Q2 OR: MILP primary; stochastic programming, CVaR, robust OR, and DP/MDP as extensions.
- Q3 Behavior: Acceptance, utilization, PD uplift, and churn jointly determine cohort economics; miscalibration rotates the frontier.
- Q4 Macro: PD/LGD/demand multipliers shrink the feasible set; maintain stress-conditioned policies.
- Q5 Innovation: Constrained bandits/RL, risk-based pricing co-moved with limits, and behavioral nudges—always measured against the MILP baseline under identical constraints.

7. Limitations

(1) Synthetic external validity unproven for any live lender portfolio. (2) Parametric elasticities need A/B offer logs. (3) Certainty-equivalent MILP; clustering/fat tails only via stress proxy. (4) Fairness not audited. (5) APR held fixed—no joint limit–price optimization. (6) Single-period; multi-period franchise value incomplete.

8. Recommendations to operationalize

1. Champion–challenger shadow vs current policy.
2. Monthly re-optimization with early-warning freezes.
3. Model-risk documentation (assumptions, drift monitors, overrides).
4. Stratified A/B within MILP-eligible segments.
5. Macro playbooks with pre-approved tier caps.
6. Offline bandit sandbox with action masks; couple limits to risk-based fees when RAROC turns negative.
7. Python/SQL feature store + reproducible batch solve (CBC/Gurobi).

9. Conclusion

A literature-aware, assumption-explicit, validated predict-then-optimize pipeline shows that constrained MILP allocation of limit-increase tiers can substantially improve expected profitability while improving—not merely tolerating—portfolio risk metrics relative to naive and unconstrained heuristics. The companion notebook makes results reproducible for assessment and adaptable once live offer and performance data are available.

Companion files: [form_answers.md](#) · [loan_limit_optimization.ipynb](#) · [loan_limit_engine.py](#) · [ml_behavioral_models.py](#) · [data/](#) · [figures/](#)